



Degree Project in Computer Science and Engineering

Second cycle, 30 credits

Adaptive Hybrid RAG

Optimizing the Cost-Performance Trade-off via Learned Query
Routing

DANTE WESSLUND

$$N = 150$$

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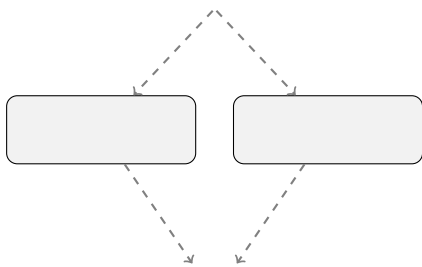
$$q\mathcal{D}_k(q)ka$$

$$p(a \mid q) = \sum_{d \in \mathcal{D}_k(q)} p_{\theta}(a \mid q, d) \, p_{\eta}(d \mid q),$$

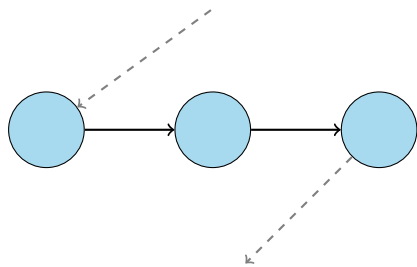
$$p_{\eta}(d \mid q)\eta p_{\theta}(a \mid q, d)\theta$$

$$X$$

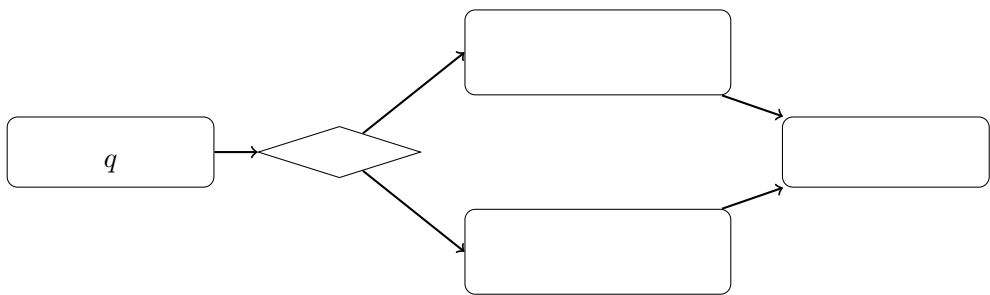
$$k$$

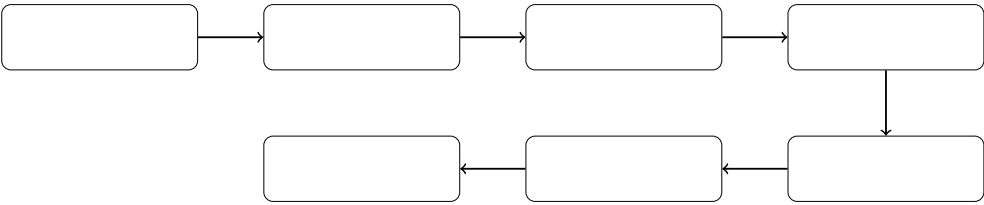


$\Rightarrow$



$\Rightarrow$





$$q_i$$

$$r_i^N=R_N(q_i), r_i^M=R_M(q_i),$$

$$\begin{array}{l} R_N R_M r_i^N r_i^M t_i^N t_i^M \\ N=150 \bar{u}^N \bar{u}^M \end{array}$$

$$\begin{array}{l} r_i^N r_i^N r_i^M \\ y_i \in \{0,1\} \end{array}$$

$$y_i=\left\{ \begin{array}{l} 0, \\ 1, \end{array} \right.$$



$$\begin{array}{c} | \\ | \\ \hline | \\ | \\ | \end{array}$$

$$q_i s_\phi(q_i) \in [0,1] \phi P_\phi(y_i = 1 \mid q_i) y_i 1 q_i$$

$$\begin{array}{l} V_i \in \mathbb{R}^V \\ s_\phi(q_i) = \sigma(\mathbb{T}_i + b), \sigma(z) = \frac{1}{1 + (-z)}, \\ \in \mathbb{R}^V b \in \mathbb{R} L_2 \end{array}$$

$$\begin{array}{l} h_i \in \mathbb{R}^d d \\ z_i = Wh_i + b, W \in \mathbb{R}^{2 \times d}, b \in \mathbb{R}^2, \end{array}$$

$$\begin{array}{l} z_{i,0} z_{i,1} \\ s_\phi(q_i) = \frac{(z_{i,1})}{(z_{i,0}) + (z_{i,1})}. \end{array}$$

$$\begin{array}{l} s_\phi(q_i) = \sigma(z_{i,1} - z_{i,0}) \\ n \\ \mathcal{L}(\phi) = -\frac{1}{n} \sum_{i=1}^n \alpha_{y_i} \, P_\phi(y_i \mid q_i), \end{array}$$

$$\alpha_c = n/(2\,n_c) c \in \{0,1\} n_c c$$

$$s_\phi(q_i)\tau\in[0,1]\pi_\tau$$

$$\pi_\tau(q)=\begin{cases} M, s_\phi(q)\geq \tau, \\ N, s_\phi(q)<\tau, \end{cases}$$

$$MNPR$$

$$F_1=\frac{2PR}{P+R}.$$

$$r_i^Nr_i^Mt_i^Nt_i^M\pi_\tau\tau\\ m\{\cdot\}10r_M(\pi_\tau)=\frac{1}{m}\sum_{i=1}^m\{\pi_\tau(q_i)=M\}\pi_\tau C(\pi_\tau)$$

$$C(\pi_\tau)=\frac{1}{m}\sum_{i=1}^m\left[\{\pi_\tau(q_i)=N\}t_i^N+\{\pi_\tau(q_i)=M\}t_i^M\right].$$

$$U(\pi_\tau)\bar{u}^N\bar{u}^M$$

$$U(\pi_\tau)=(1-r_M(\pi_\tau))\,\bar{u}^N+r_M(\pi_\tau)\,\bar{u}^M.$$

$$(\pi_\tau)=\frac{1}{m}\sum_{i=1}^m\{y_i=1,\,\pi_\tau(q_i)=N\},$$

$$(\pi_\tau)=\frac{1}{m}\sum_{i=1}^m\{y_i=0,\,\pi_\tau(q_i)=M\}.$$

$$220,000L_2C=1.01,000$$

$$2\times 10^{-5}0.0181610\%1.0\alpha_c833$$

$$71342$$

$$\tau^*[0.05,0.95]\tau^*$$

$$50\%$$

$$95\%1\,000\pm$$

$$\pi_{\tau}\tau$$

$\tau_{010.01}$

$0.078 \pm 0.0190.1550.0570.076 \pm 0.009$

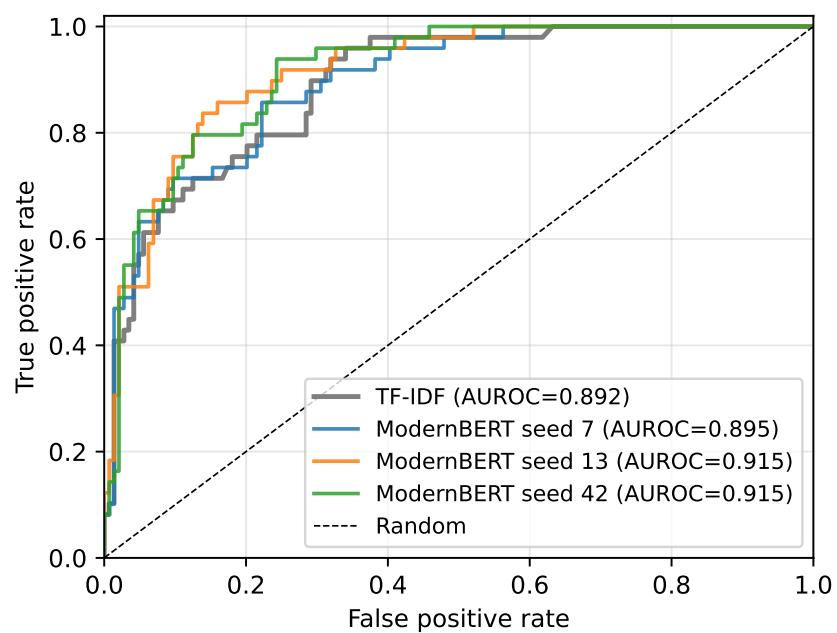
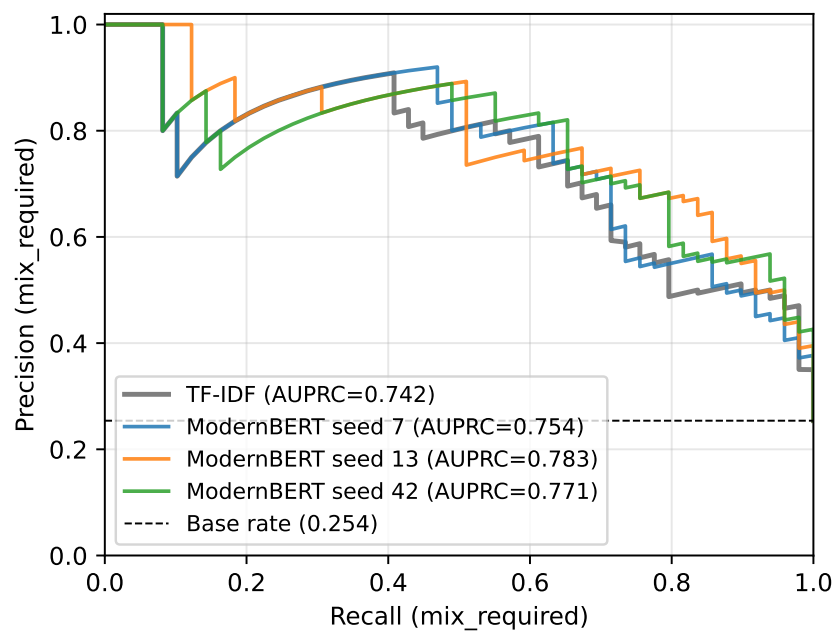
$0.6990.6670.7010.5850.8460.8030.7768,5317,13817,85752 \%60 \%$

0.5000.25495 %1 000±

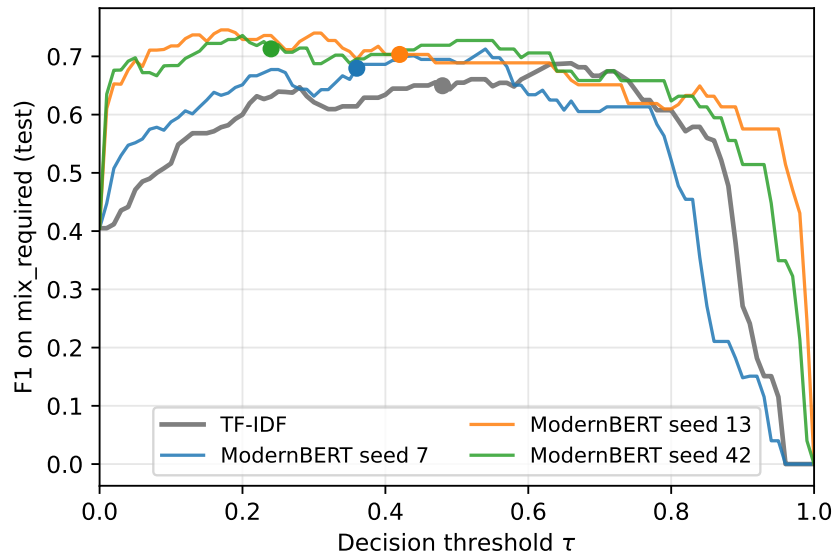
0.769 ± 0.015 0.908 ± 0.011 0.798 ± 0.012 0.701 ± 0.057 0.701 ± 0.042 0.699 ± 0.017 0.846 ± 0.016

$$N = 150 \pm$$

$$0.256 \pm 0.027 \quad 14.92 \pm 0.69 \quad 0.076 \pm 0.009 \quad 0.078 \pm 0.019$$

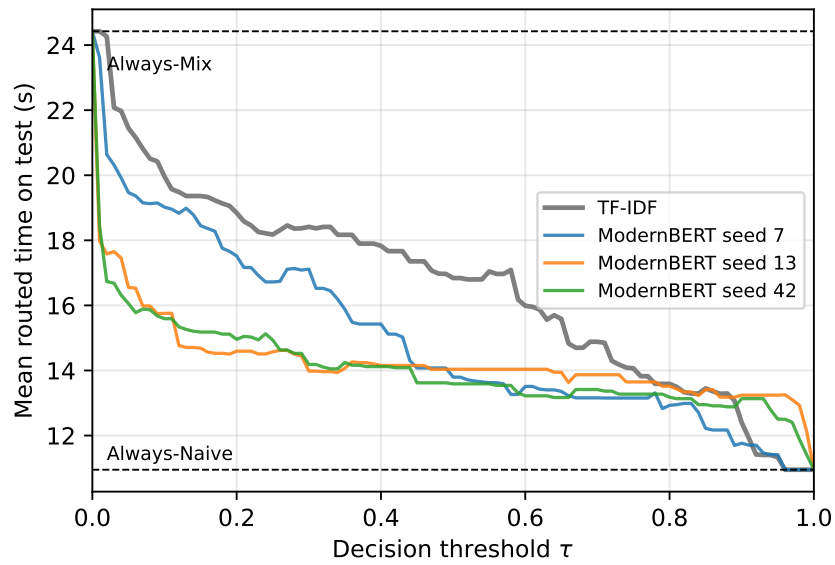


$$\tau^* = 0.24, 0.36, 0.42427130.6990.017$$

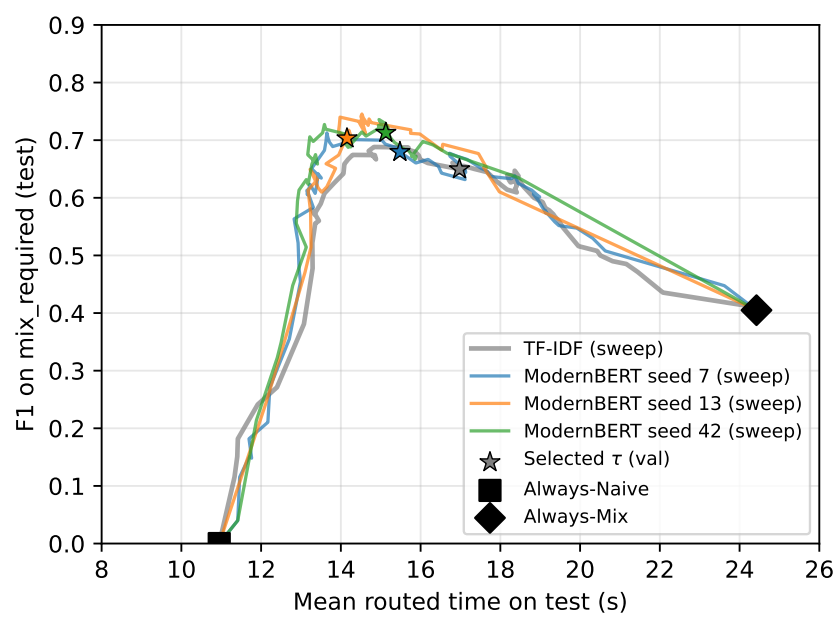


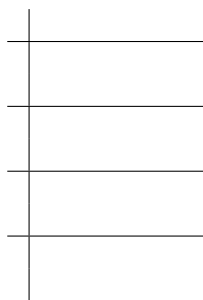
$$\tau^* = 0.4842 \tau^* = 0.247 \tau^* = 0.3613 \tau^* = 0.42$$

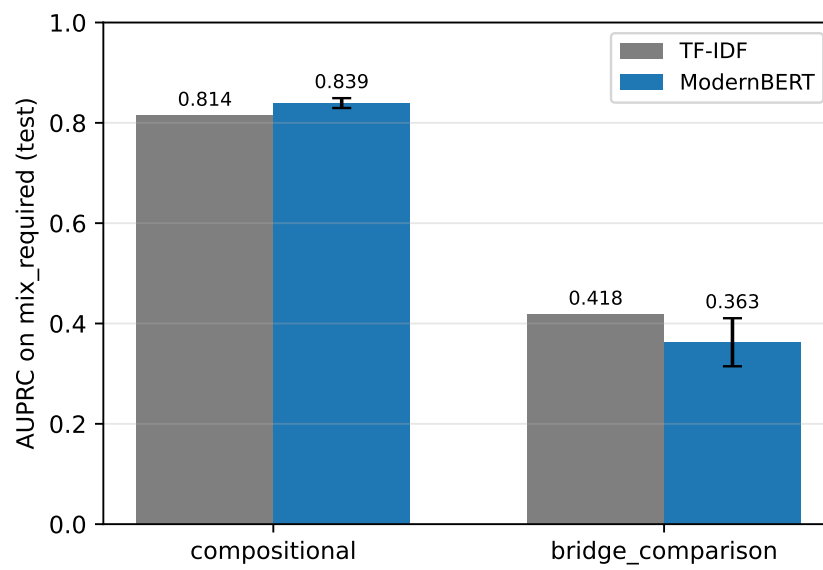
$$01\tau = 0\tau = 1$$



τ
2908333865150.8390.8140.7990.7530.3750.1670.4180.3630.3160.2110.345
0.2730.7070.606







0.8390.8140.7530.7990.4180.316
0.667

$193[0.836, 0.936]0.908193 \leq 0.0170.0570.042193$

$10.9524.423,45817,85715173039 \%7,1008,5005260 \%$

$$1930.7690.7420.7980.7840.6990.6500.8460.788 \leq 0.0170.0570.076 \pm 0.0090.078 \pm 0.0190.155$$

$$151710.9524.427,1008,5003,45817,85760 \%52 \%$$

$$0.6670.7940.803$$

$$N = 1501,286$$

$$1,286$$

